TECNOLÓGICO DE ESTUDIOS SUPERIORES DE VILLA GUERRERO.

INGENIERÍA EN SISTEMAS COMPUTACIONALES

UNIDAD DE APRENDIZAJE:

SISTEMAS OPERATIVOS

PRACTICA 1

ADMINISTRACION Y MONITOREO DE PROCESOS EN LINUX

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PRÁCTICA No. 1: ADMINISTRACIÓN Y MONITOREO DE PROCESOS EN LINUX

Introducción

En esta práctica se utilizaron diferentes comandos de Linux para observar, identificar y analizar los procesos que se encuentran ejecutándose en el sistema operativo. Los procesos son programas o tareas que están siendo ejecutados por el sistema, y conocer su estado permite comprender mejor el funcionamiento de Linux y administrar los recursos del equipo.

Para realizar la práctica se utilizaron principalmente los comandos **`ps`**, **`ps -ef`**, **`grep`**, **`ps -u user`** y diferentes formatos de **`ps -eo`**, con los cuales fue posible consultar información como el identificador de proceso (PID), usuario, proceso padre, tiempo de ejecución, comando y porcentaje de CPU utilizado.

Objetivo

Identificar y analizar los procesos activos de un sistema operativo Linux mediante el uso de comandos de terminal, comprendiendo la información que proporciona cada uno y su utilidad para la administración y monitoreo del sistema.

Desarrollo

1. Comando `ps`

El comando **`ps`** permite visualizar los procesos asociados a la terminal actual. Al ejecutarlo sin opciones, muestra principalmente el **PID**, la terminal, el tiempo de ejecución y el comando correspondiente.

En la captura se observan dos procesos principales:

- **PID 3927:** corresponde a la terminal `bash`.
- **PID 4082:** corresponde al propio comando `ps` que se está ejecutando.

Esto demuestra que `ps`, al utilizarse de manera sencilla, muestra principalmente los procesos relacionados con la sesión actual.

Captura 1. Ejecución del comando `ps`

```
user@debian:~$ ps
    PID TTY          TIME CMD
   3927 pts/0    00:00:00 bash
   4082 pts/0    00:00:00 ps
```

---

2. Comando `ps -ef`

El comando `ps -ef` permite obtener una vista más completa de los procesos que se están ejecutando en el sistema.

Las principales columnas que aparecen son:

- **UID:** usuario que ejecuta el proceso.
- **PID:** identificador único del proceso.
- **PPID:** identificador del proceso padre.
- **C:** utilización de CPU.
- **STIME:** hora en la que inició el proceso.
- **TTY:** terminal asociada.
- **TIME:** tiempo de CPU utilizado.
- **CMD:** comando utilizado para iniciar el proceso.

En las capturas se observa una gran cantidad de procesos pertenecientes al usuario **root**, incluyendo procesos del kernel, servicios del sistema, controladores y procesos relacionados con el entorno gráfico.

También se observan procesos pertenecientes al usuario **user**, como `gnome-shell`, `firefox-esr`, `gnome-terminal` y diferentes servicios del entorno de escritorio.

Un aspecto importante es la relación entre **PID** y **PPID**, ya que permite conocer qué proceso creó o inició a otro proceso.

Capturas 2-5. Ejecución del comando `ps -ef`

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```
user@debian:~$ ps -ef
UID          PID    PPID  C STIME TTY          TIME CMD
root           1       0  0 18:48 ?        00:00:01 /sbin/init components splash
root           2       0  0 18:48 ?        00:00:00 [kthreadd]
root           3       2  0 18:48 ?        00:00:00 [pool_workqueue_release]
root           4       2  0 18:48 ?        00:00:00 [kworker/R-kvfree_rcu_reclai
root           5       2  0 18:48 ?        00:00:00 [kworker/R-rcu_gp]
root           6       2  0 18:48 ?        00:00:00 [kworker/R-sync_wq]
root           7       2  0 18:48 ?        00:00:00 [kworker/R-slub_flushwq]
root           8       2  0 18:48 ?        00:00:00 [kworker/R-netns]
root           9       2  0 18:48 ?        00:00:00 [kworker/0:0-events]
root          10       2  0 18:48 ?        00:00:00 [kworker/0:0H-events_highpri
root          13       2  0 18:48 ?        00:00:00 [kworker/R-mm_percpu_wq]
root          14       2  0 18:48 ?        00:00:00 [rcu_tasks_kthread]
root          15       2  0 18:48 ?        00:00:00 [rcu_tasks_rude_kthread]
root          16       2  0 18:48 ?        00:00:00 [rcu_tasks_trace_kthread]
root          17       2  0 18:48 ?        00:00:00 [ksoftirqd/0]
root          18       2  0 18:48 ?        00:00:00 [rcu_preempt]
root          19       2  0 18:48 ?        00:00:00 [rcu_exp_par_gp_kthread_work
root          20       2  0 18:48 ?        00:00:00 [rcu_exp_gp_kthread_worker]
root          21       2  0 18:48 ?        00:00:00 [migration/0]
root          22       2  0 18:48 ?        00:00:00 [idle_inject/0]
root          23       2  0 18:48 ?        00:00:00 [cpuhp/0]
root          24       2  0 18:48 ?        00:00:00 [cpuhp/1]
root          25       2  0 18:48 ?        00:00:00 [idle_inject/1]
root          26       2  0 18:48 ?        00:00:00 [migration/1]
root          27       2  0 18:48 ?        00:00:00 [ksoftirqd/1]
root          29       2  0 18:48 ?        00:00:00 [kworker/1:0H-events_highpri
root          30       2  0 18:48 ?        00:00:00 [cpuhp/2]
root          31       2  0 18:48 ?        00:00:00 [idle_inject/2]
root          32       2  0 18:48 ?        00:00:00 [migration/2]
root          33       2  0 18:48 ?        00:00:00 [ksoftirqd/2]
root          35       2  0 18:48 ?        00:00:00 [kworker/2:0H-events_highpri
root          36       2  0 18:48 ?        00:00:00 [cpuhp/3]
root          37       2  0 18:48 ?        00:00:00 [idle_inject/3]
root          38       2  0 18:48 ?        00:00:00 [migration/3]
root          39       2  0 18:48 ?        00:00:00 [ksoftirqd/3]
root          40       2  0 18:48 ?        00:00:00 [kworker/3:0-cgroup_free]
root          41       2  0 18:48 ?        00:00:00 [kworker/3:0H-events_highpri
root          42       2  0 18:48 ?        00:00:00 [cpuhp/4]
root          43       2  0 18:48 ?        00:00:00 [idle_inject/4]
root          44       2  0 18:48 ?        00:00:00 [migration/4]
root          45       2  0 18:48 ?        00:00:00 [ksoftirqd/4]
root          47       2  0 18:48 ?        00:00:00 [kworker/4:0H-events_highpri
root          48       2  0 18:48 ?        00:00:00 [cpuhp/5]
root          49       2  0 18:48 ?        00:00:00 [idle_inject/5]
root          50       2  0 18:48 ?        00:00:00 [migration/5]
root          51       2  0 18:48 ?        00:00:00 [ksoftirqd/5]
root          53       2  0 18:48 ?        00:00:00 [kworker/5:0H-events_highpri
root          54       2  0 18:48 ?        00:00:00 [cpuhp/6]
root          55       2  0 18:48 ?        00:00:00 [idle_inject/6]
root          56       2  0 18:48 ?        00:00:00 [migration/6]
root          57       2  0 18:48 ?        00:00:00 [ksoftirqd/6]
root          58       2  0 18:48 ?        00:00:00 [kworker/6:0-mm_percpu_wq]
```

```
root          59       2  0 18:48 ?        00:00:00 [kworker/6:0H-events_highpri
root          60       2  0 18:48 ?        00:00:00 [cpuhp/7]
root          61       2  0 18:48 ?        00:00:00 [idle_inject/7]
root          62       2  0 18:48 ?        00:00:00 [migration/7]
root          63       2  0 18:48 ?        00:00:00 [ksoftirqd/7]
root          65       2  0 18:48 ?        00:00:00 [kworker/7:0H-events_highpri
root          69       2  0 18:48 ?        00:00:00 [kworker/u32:3-async]
root          72       2  0 18:48 ?        00:00:00 [kworker/u32:6-events_unboun
root          73       2  0 18:48 ?        00:00:00 [kworker/u32:7-i915]
root          74       2  0 18:48 ?        00:00:00 [kdevtmpfs]
root          75       2  0 18:48 ?        00:00:00 [kworker/R-inet_frag_wq]
root          76       2  0 18:48 ?        00:00:00 [kauditd]
root          77       2  0 18:48 ?        00:00:00 [khungtaskd]
root          78       2  0 18:48 ?        00:00:00 [oom_reaper]
root          79       2  0 18:48 ?        00:00:00 [kworker/R-writeback]
root          80       2  0 18:48 ?        00:00:00 [kcompactd0]
root          81       2  0 18:48 ?        00:00:00 [ksmd]
root          82       2  0 18:48 ?        00:00:00 [khugepaged]
root          83       2  0 18:48 ?        00:00:00 [kworker/R-kintegrityd]
root          84       2  0 18:48 ?        00:00:00 [kworker/R-kblockd]
root          85       2  0 18:48 ?        00:00:00 [kworker/R-blkcg_punt_bio]
root          86       2  0 18:48 ?        00:00:00 [irq/9-acpi]
root          87       2  0 18:48 ?        00:00:00 [kworker/2:1-events]
root          88       2  0 18:48 ?        00:00:00 [kworker/6:1-events_freezabl
root          89       2  0 18:48 ?        00:00:00 [kworker/4:1-events]
root          90       2  0 18:48 ?        00:00:00 [kworker/1:1-events]
root          91       2  0 18:48 ?        00:00:00 [kworker/5:1-events]
root          92       2  0 18:48 ?        00:00:00 [kworker/R-tpm_dev_wq]
root          93       2  0 18:48 ?        00:00:00 [kworker/R-edac-poller]
root          94       2  0 18:48 ?        00:00:00 [kworker/R-devfreq_wq]
root          95       2  0 18:48 ?        00:00:00 [kworker/R-quota_events_unbo
root          96       2  0 18:48 ?        00:00:00 [kworker/0:1H-kblockd]
root          97       2  0 18:48 ?        00:00:00 [kswapd0]
root          99       2  0 18:48 ?        00:00:00 [kworker/7:1-i915-unordered]
root         106       2  0 18:48 ?        00:00:00 [kworker/R-kthrotld]
root         110       2  0 18:48 ?        00:00:00 [kworker/R-acpi_thermal_pm]
root         111       2  0 18:48 ?        00:00:00 [hwrng]
root         112       2  0 18:48 ?        00:00:00 [kworker/R-mld]
root         113       2  0 18:48 ?        00:00:00 [kworker/R-ipv6_addrconf]
root         118       2  0 18:48 ?        00:00:00 [kworker/R-kstrp]
root         120       2  0 18:48 ?        00:00:00 [kworker/u33:0-i915_flip]
root         175       2  0 18:48 ?        00:00:00 [kworker/5:1H-kblockd]
root         208       2  0 18:48 ?        00:00:00 [kworker/7:1H-events_highpri
root         209       2  0 18:48 ?        00:00:00 [kworker/7:2-events]
root         218       2  0 18:48 ?        00:00:00 [kworker/2:1H-kblockd]
root         227       2  0 18:48 ?        00:00:00 [kworker/6:1H-events_highpri
root         228       2  0 18:48 ?        00:00:00 [kworker/1:1H-events_highpri
root         237       2  0 18:48 ?        00:00:00 [kworker/4:1H-events_highpri
root         256       2  0 18:48 ?        00:00:00 [kworker/3:1H-kblockd]
root         257       2  0 18:48 ?        00:00:00 [kworker/4:2-mm_percpu_wq]
root         259       2  0 18:48 ?        00:00:00 [kworker/1:2-events]
root         261       2  0 18:48 ?        00:00:00 [kworker/5:2-events_freezabl
root         262       2  0 18:48 ?        00:00:00 [watchdogd]
root         263       2  0 18:48 ?        00:00:00 [irq/144-MSFT0001:01]
```

```
root         264       2  0 18:48 ?        00:00:00 [kworker/R-nvme-wq]
root         265       2  0 18:48 ?        00:00:00 [kworker/R-nvme-reset-wq]
root         266       2  0 18:48 ?        00:00:00 [kworker/R-nvme-delete-wq]
root         267       2  0 18:48 ?        00:00:00 [kworker/R-nvme-auth-wq]
root         268       2  0 18:48 ?        00:00:00 [kworker/R-ata_sff]
root         269       2  0 18:48 ?        00:00:00 [irq/143-GTCH7503:00]
root         270       2  0 18:48 ?        00:00:00 [scsi_eh_0]
root         271       2  0 18:48 ?        00:00:00 [kworker/R-scsi_tmf_0]
root         272       2  0 18:48 ?        00:00:00 [scsi_eh_1]
root         273       2  0 18:48 ?        00:00:00 [kworker/R-scsi_tmf_1]
root         274       2  0 18:48 ?        00:00:00 [kworker/3:2-mm_percpu_wq]
root         277       2  0 18:48 ?        00:00:00 [spi0]
root         278       2  0 18:48 ?        00:00:00 [kworker/0:2-events]
root         280       2  0 18:48 ?        00:00:00 [scsi_eh_2]
root         281       2  0 18:48 ?        00:00:00 [kworker/R-scsi_tmf_2]
root         282       2  0 18:48 ?        00:00:00 [usb-storage]
root         283       2  0 18:48 ?        00:00:00 [kworker/R-ttm]
root         284       2  0 18:48 ?        00:00:00 [card0-crtc0]
root         285       2  0 18:48 ?        00:00:00 [card0-crtc1]
root         286       2  0 18:48 ?        00:00:00 [card0-crtc2]
root         287       2  0 18:48 ?        00:00:00 [card0-crtc3]
root         288       2  0 18:48 ?        00:00:00 [kworker/R-uas]
root         300       2  0 18:48 ?        00:00:00 [kworker/R-md]
root         301       2  0 18:48 ?        00:00:00 [kworker/R-md_bitmap]
root         307       2  0 18:48 ?        00:00:00 [kworker/R-raid5wq]
root         485       2  0 18:48 ?        00:00:00 [psimon]
root         499       1  0 18:48 ?        00:00:00 /usr/lib/systemd/systemd-jou
root         547       2  0 18:48 ?        00:00:00 [kworker/2:2-events]
root         551       1  0 18:48 ?        00:00:00 /usr/lib/systemd/systemd-ude
root         556       2  0 18:48 ?        00:00:00 [psimon]
root         621       2  0 18:48 ?        00:00:00 [irq/172-mei_me]
root         637       2  0 18:48 ?        00:00:00 [kworker/R-cryptd]
root         638       2  0 18:48 ?        00:00:00 [kworker/R-cfg80211]
root         677       2  0 18:48 ?        00:00:00 [kworker/u32:10-events_unbou
root         681       2  0 18:48 ?        00:00:00 [kworker/u32:12-events_unbou
root         683       2  0 18:48 ?        00:00:00 [kworker/u32:13-events_unbou
root         779       2  0 18:48 ?        00:00:00 [kworker/u33:1-rtw_tx_wq]
root         835       2  1 18:48 ?        00:00:05 [irq/173-rtw88_pci]
root         915       2  0 18:48 ?        00:00:00 [irq/174-AudioDSP]
root        1452       1  0 18:49 ?        00:00:00 /usr/libexec/accounts-daemon
avahi       1455       1  0 18:49 ?        00:00:00 avahi-daemon: running [debia
root        1456       1  0 18:49 ?        00:00:00 /usr/libexec/bluetooth/bluet
root        1457       1  0 18:49 ?        00:00:00 /usr/sbin/cron -f
message+    1458       1  0 18:49 ?        00:00:00 /usr/bin/dbus-daemon --syste
root        1464       1  0 18:49 ?        00:00:00 /usr/libexec/low-memory-moni
polkitd     1465       1  0 18:49 ?        00:00:00 /usr/lib/polkit-1/polkitd --
root        1466       1  0 18:49 ?        00:00:00 /usr/sbin/smartd -n
root        1467       1  0 18:49 ?        00:00:00 /usr/libexec/switcheroo-cont
root        1469       1  0 18:49 ?        00:00:00 /usr/lib/systemd/systemd-log
root        1470       1  0 18:49 ?        00:00:00 /usr/libexec/udisks2/udisksd
avahi       1471    1455  0 18:49 ?        00:00:00 avahi-daemon: chroot helper
root        1511       1  0 18:49 ?        00:00:01 /usr/sbin/NetworkManager --n
root        1512       1  0 18:49 ?        00:00:00 /usr/sbin/wpa_supplicant -u
root        1522       2  0 18:49 ?        00:00:00 [psimon]
```

```
root        1546       1  0 18:49 ?        00:00:00 /usr/sbin/ModemManager
root        1560       2  0 18:49 ?        00:00:00 [krfcommd]
root        1575       1  0 18:49 ?        00:00:00 /usr/sbin/cupsd -l
root        1593       1  0 18:49 ?        00:00:00 /usr/sbin/gdm3
root        1599    1593  0 18:49 ?        00:00:00 gdm-session-worker [pam/gdm-
user        1607       1  0 18:49 ?        00:00:00 /usr/lib/systemd/systemd --u
user        1609    1607  0 18:49 ?        00:00:00 (sd-pam)
user        1629    1607  0 18:49 ?        00:00:00 /usr/bin/dbus-daemon --sessi
user        1631    1607  0 18:49 ?        00:00:00 /usr/bin/pipewire
user        1633    1607  0 18:49 ?        00:00:00 /usr/bin/pipewire -c filter-
user        1634    1607  0 18:49 ?        00:00:00 /usr/bin/wireplumber
user        1635    1607  0 18:49 ?        00:00:00 /usr/bin/pipewire-pulse
user        1637    1607  0 18:49 ?        00:00:00 /usr/bin/gnome-keyring-daemo
user        1638    1607  0 18:49 ?        00:00:00 /usr/bin/mpris-proxy
rtkit       1647       1  0 18:49 ?        00:00:00 /usr/libexec/rtkit-daemon
user        1650    1599  0 18:49 tty2     00:00:00 /usr/libexec/gdm-wayland-ses
user        1659    1650  0 18:49 tty2     00:00:00 /usr/libexec/gnome-session-b
user        1709    1607  0 18:49 ?        00:00:00 /usr/libexec/gcr-ssh-agent -
user        1710    1607  0 18:49 ?        00:00:00 /usr/libexec/gnome-session-c
user        1711    1607  0 18:49 ?        00:00:00 /usr/bin/ssh-agent -D
user        1721    1607  0 18:49 ?        00:00:00 /usr/libexec/gvfsd
user        1729    1607  0 18:49 ?        00:00:00 /usr/libexec/gvfsd-fuse /run
user        1731    1607  0 18:49 ?        00:00:00 /usr/libexec/gnome-session-b
user        1765    1731  0 18:49 ?        00:00:00 /usr/libexec/at-spi-bus-laun
user        1783    1765  0 18:49 ?        00:00:00 /usr/bin/dbus-daemon --confi
user        1788    1607  4 18:49 ?        00:00:20 /usr/bin/gnome-shell
user        1809    1607  0 18:49 ?        00:00:00 /usr/libexec/xdg-permission-
root        1817       1  0 18:49 ?        00:00:00 /usr/sbin/cups-browsed
user        1840    1607  0 18:49 ?        00:00:00 /usr/libexec/at-spi2-registr
colord      1841       1  0 18:49 ?        00:00:00 /usr/libexec/colord
user        1877    1607  0 18:49 ?        00:00:00 /usr/libexec/gnome-shell-cal
user        1885    1607  0 18:49 ?        00:00:00 /usr/libexec/dconf-service
user        1888    1607  0 18:49 ?        00:00:00 /usr/libexec/evolution-sourc
user        1890    1607  0 18:49 ?        00:00:00 /usr/bin/gjs -m /usr/share/g
root        1910       1  0 18:49 ?        00:00:00 /usr/libexec/upowerd
user        1911    1607  0 18:49 ?        00:00:00 /usr/bin/ibus-daemon --panel
user        1916    1607  0 18:49 ?        00:00:00 /usr/libexec/gsd-a11y-settin
user        1917    1607  0 18:49 ?        00:00:00 /usr/libexec/gsd-color
user        1918    1607  0 18:49 ?        00:00:00 /usr/libexec/gsd-datetime
user        1921    1607  0 18:49 ?        00:00:00 /usr/libexec/gsd-housekeepin
user        1924    1607  0 18:49 ?        00:00:00 /usr/libexec/gsd-keyboard
user        1926    1607  0 18:49 ?        00:00:00 /usr/libexec/gsd-media-keys
user        1928    1607  0 18:49 ?        00:00:00 /usr/libexec/gsd-power
user        1929    1607  0 18:49 ?        00:00:00 /usr/libexec/gsd-print-notif
user        1930    1607  0 18:49 ?        00:00:00 /usr/libexec/gsd-rfkill
user        1931    1607  0 18:49 ?        00:00:00 /usr/libexec/gsd-screensaver
user        1934    1607  0 18:49 ?        00:00:00 /usr/libexec/gsd-sharing
user        1936    1607  0 18:49 ?        00:00:00 /usr/libexec/gsd-smartcard
user        1937    1607  0 18:49 ?        00:00:00 /usr/libexec/gsd-sound
user        1939    1607  0 18:49 ?        00:00:00 /usr/libexec/gsd-usb-protect
user        1940    1607  0 18:49 ?        00:00:00 /usr/libexec/gsd-wacom
user        1957    1607  0 18:49 ?        00:00:00 /usr/libexec/gvfs-udisks2-vo
user        2014    1607  0 18:49 ?        00:00:00 /usr/bin/gjs -m /usr/share/g
user        2018    1607  0 18:49 ?        00:00:00 /usr/libexec/gvfs-mtp-volume
```

```
user        2018    1607  0 18:49 ?        00:00:00 /usr/libexec/gvfs-mtp-volume
user        2043    1607  0 18:49 ?        00:00:00 /usr/libexec/gsd-printer
user        2048    1607  0 18:49 ?        00:00:00 /usr/libexec/gvfs-gphoto2-vo
user        2070    1731  0 18:49 ?        00:00:00 /usr/libexec/gsd-disk-utilit
user        2078    1731  2 18:49 ?        00:00:13 /usr/bin/gnome-software --ga
user        2081    1731  0 18:49 ?        00:00:00 /usr/libexec/evolution-data-
user        2150    1607  0 18:49 ?        00:00:00 /usr/libexec/gvfs-goa-volume
user        2159    1607  0 18:49 ?        00:00:00 /usr/libexec/goa-daemon
user        2171    1607  0 18:49 ?        00:00:00 /usr/libexec/goa-identity-se
user        2174    1607  0 18:49 ?        00:00:00 /usr/libexec/gvfs-afc-volume
user        2185    1788  0 18:49 ?        00:00:00 /usr/bin/Xwayland :0 -rootle
user        2249    1607  0 18:49 ?        00:00:00 /usr/libexec/xdg-desktop-por
user        2254    1607  0 18:49 ?        00:00:00 /usr/libexec/xdg-document-po
user        2257    1607  0 18:49 ?        00:00:00 /usr/libexec/evolution-calen
root        2263    2254  0 18:49 ?        00:00:00 fusermount3 -o rw,nosuid,nod
user        2285    1607  0 18:49 ?        00:00:00 /usr/libexec/gvfsd-metadata
user        2292    1607  0 18:49 ?        00:00:00 /usr/libexec/gsd-xsettings
user        2297    1607  0 18:49 ?        00:00:00 /usr/libexec/evolution-addre
user        2321    1788  0 18:49 ?        00:00:00 /usr/libexec/mutter-x11-fram
user        2392    1911  0 18:49 ?        00:00:00 /usr/libexec/ibus-dconf
user        2393    1911  0 18:49 ?        00:00:01 /usr/libexec/ibus-extension-
user        2399    1607  0 18:49 ?        00:00:00 /usr/libexec/ibus-portal
user        2402    1607  0 18:49 ?        00:00:00 /usr/libexec/localsearch-3
user        2403    1607  0 18:49 ?        00:00:00 /usr/libexec/xdg-desktop-por
user        2422    1911  0 18:49 ?        00:00:00 /usr/lib/ibus-mozc/ibus-engi
user        2430    2422  0 18:49 ?        00:00:00 /usr/lib/mozc/mozc_server
user        2465    1607  0 18:49 ?        00:00:00 /usr/libexec/xdg-desktop-por
root        2506       1  0 18:49 ?        00:00:03 /usr/libexec/fwupd/fwupd
user        2592    1911  0 18:49 ?        00:00:00 /usr/libexec/ibus-engine-sim
root        2599       1  0 18:49 ?        00:00:00 /usr/libexec/power-profiles-
user        2707    1788 21 18:49 ?        00:01:33 /usr/lib/firefox-esr/firefox
user        2719    1607  0 18:49 ?        00:00:00 /usr/lib/firefox-esr/crashhe
user        2795    2707  0 18:49 ?        00:00:00 /usr/lib/firefox-esr/firefox
user        2856    2707  0 18:49 ?        00:00:00 /usr/lib/firefox-esr/firefox
user        2872    2707  0 18:49 ?        00:00:00 /usr/lib/firefox-esr/firefox
user        2895    2707  0 18:49 ?        00:00:01 /usr/lib/firefox-esr/firefox
user        2950    2707  0 18:49 ?        00:00:00 /usr/lib/firefox-esr/firefox
user        3027    2707  0 18:49 ?        00:00:00 /usr/lib/firefox-esr/firefox
user        3041    2707 47 18:49 ?        00:03:24 /usr/lib/firefox-esr/firefox
root        3112       2  0 18:49 ?        00:00:00 [kworker/u33:3-rtw_tx_wq]
lp          3186    1575  0 18:49 ?        00:00:00 /usr/lib/cups/notifier/dbus
lp          3187    1575  0 18:49 ?        00:00:00 /usr/lib/cups/notifier/dbus
user        3582    2707  0 18:50 ?        00:00:00 /usr/lib/firefox-esr/firefox
user        3630    2707  0 18:50 ?        00:00:00 /usr/lib/firefox-esr/firefox
root        3839       2  0 18:54 ?        00:00:00 [kworker/1:0-i915-unordered]
root        3849       2  0 18:54 ?        00:00:00 [kworker/0:1-events]
root        3869       2  0 18:55 ?        00:00:00 [kworker/4:0]
user        3919    1607  0 18:55 ?        00:00:00 /usr/libexec/gnome-terminal-
user        3927    3919  0 18:55 pts/0    00:00:00 bash
root        3958       2  0 18:55 ?        00:00:00 [kworker/5:0]
root        4022       2  0 18:55 ?        00:00:00 [kworker/7:0-i915-unordered]
user        4023    2707  0 18:55 ?        00:00:00 /usr/lib/firefox-esr/firefox
root        4056       2  0 18:56 ?        00:00:00 [kworker/3:1]
user        4117    3927  0 18:57 pts/0    00:00:00 ps -ef
```

---

3. Búsqueda de Firefox mediante `grep`

Para localizar específicamente los procesos relacionados con Firefox se utilizó el comando:

```
ps -ef | grep firefox
```

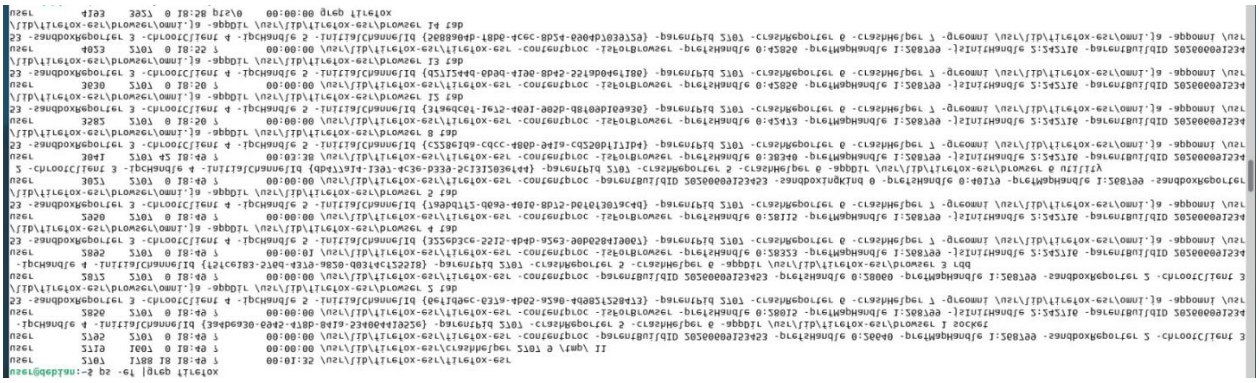
El operador `|` permite enviar la información generada por `ps -ef` al comando `grep`. Posteriormente, `grep firefox` filtra los resultados y muestra únicamente las líneas que contienen la palabra **firefox**.

En las capturas se pueden observar diferentes procesos relacionados con **Firefox ESR**, entre ellos el proceso principal y varios procesos secundarios, como:

- `firefox-esr`
- `WebExtensions`
- `Socket Process`
- `RDD Process`
- `Privileged Content`
- `Utility Process`
- `Isolated Web Content`

Esto permite comprobar que una aplicación como Firefox no necesariamente corresponde a un solo proceso, sino que puede utilizar varios procesos independientes para realizar diferentes tareas.

Captura 6. Búsqueda de procesos de Firefox



---

4. Procesos del usuario con `ps -u user`

El comando:

```
ps -u user
```

permite mostrar los procesos pertenecientes específicamente al usuario **user**.

En la captura se observan procesos relacionados con el entorno gráfico y las aplicaciones que se encuentran abiertas. Entre ellos aparecen servicios de GNOME, PipeWire, D-Bus, Firefox y la terminal.

También se puede observar que algunos procesos tienen una relación jerárquica con otros. Por ejemplo, Firefox tiene un proceso principal y diferentes procesos secundarios asociados.

Este comando resulta útil cuando se desea conocer qué procesos están siendo ejecutados por un usuario determinado sin mostrar todos los procesos del sistema.

Capturas 7-8. Procesos pertenecientes al usuario `user`

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```
user@debian:~$ ps -u user
    PID TTY          TIME CMD
   1607 ?        00:00:00 systemd
   1609 ?        00:00:00 (sd-pam)
   1629 ?        00:00:00 dbus-daemon
   1631 ?        00:00:00 pipewire
   1633 ?        00:00:00 pipewire
   1634 ?        00:00:00 wireplumber
   1635 ?        00:00:00 pipewire-pulse
   1637 ?        00:00:00 gnome-keyring-d
   1638 ?        00:00:00 mpris-proxy
   1650 tty2     00:00:00 gdm-wayland-ses
   1659 tty2     00:00:00 gnome-session-b
   1709 ?        00:00:00 gcr-ssh-agent
   1710 ?        00:00:00 gnome-session-c
   1711 ?        00:00:00 ssh-agent
   1721 ?        00:00:00 gvfsd
   1729 ?        00:00:00 gvfsd-fuse
   1731 ?        00:00:00 gnome-session-b
   1765 ?        00:00:00 at-spi-bus-laun
   1783 ?        00:00:00 dbus-daemon
   1788 ?        00:00:23 gnome-shell
   1809 ?        00:00:00 xdg-permission-
   1840 ?        00:00:00 at-spi2-registr
   1877 ?        00:00:00 gnome-shell-cal
   1885 ?        00:00:00 dconf-service
   1888 ?        00:00:00 evolution-sourc
   1890 ?        00:00:00 gjs
   1911 ?        00:00:00 ibus-daemon
   1916 ?        00:00:00 gsd-a11y-settin
   1917 ?        00:00:00 gsd-color
   1918 ?        00:00:00 gsd-datetime
   1921 ?        00:00:00 gsd-housekeepin
   1924 ?        00:00:00 gsd-keyboard
   1926 ?        00:00:00 gsd-media-keys
   1928 ?        00:00:00 gsd-power
   1929 ?        00:00:00 gsd-print-notif
   1930 ?        00:00:00 gsd-rfkill
   1931 ?        00:00:00 gsd-screensaver
   1934 ?        00:00:00 gsd-sharing
   1936 ?        00:00:00 gsd-smartcard
   1937 ?        00:00:00 gsd-sound
   1939 ?        00:00:00 gsd-usb-protect
   1940 ?        00:00:00 gsd-wacom
   1957 ?        00:00:00 gvfs-udisks2-vo
   2014 ?        00:00:00 gjs
   2018 ?        00:00:00 gvfs-mtp-volume
   2043 ?        00:00:00 gsd-printer
   2048 ?        00:00:00 gvfs-gphoto2-vo
   2070 ?        00:00:00 gsd-disk-utilit
   2078 ?        00:00:13 gnome-software
   2081 ?        00:00:00 evolution-alarm
   2150 ?        00:00:00 gvfs-goa-volume
   2159 ?        00:00:00 goa-daemon
```

```
2159 ?        00:00:00 goa-daemon
2171 ?        00:00:00 goa-identity-se
2174 ?        00:00:00 gvfs-afc-volume
2185 ?        00:00:00 Xwayland
2249 ?        00:00:00 xdg-desktop-por
2254 ?        00:00:00 xdg-document-po
2257 ?        00:00:00 evolution-calen
2285 ?        00:00:00 gvfsd-metadata
2292 ?        00:00:00 gsd-xsettings
2297 ?        00:00:00 evolution-addre
2321 ?        00:00:00 mutter-x11-fram
2392 ?        00:00:00 ibus-dconf
2393 ?        00:00:01 ibus-extension-
2399 ?        00:00:00 ibus-portal
2402 ?        00:00:00 localsearch-3
2403 ?        00:00:00 xdg-desktop-por
2422 ?        00:00:00 ibus-engine-moz
2430 ?        00:00:00 mozc_server
2465 ?        00:00:00 xdg-desktop-por
2592 ?        00:00:00 ibus-engine-sim
2707 ?        00:01:39 firefox-esr
2719 ?        00:00:00 crashhelper
2795 ?        00:00:00 Socket Process
2856 ?        00:00:00 WebExtensions
2872 ?        00:00:00 RDD Process
2895 ?        00:00:01 Privileged Cont
2950 ?        00:00:01 Isolated Web Co
3027 ?        00:00:00 Utility Process
3041 ?        00:03:59 Isolated Web Co
3582 ?        00:00:00 Web Content
3630 ?        00:00:00 Web Content
3919 ?        00:00:02 gnome-terminal-
3927 pts/0    00:00:00 bash
4023 ?        00:00:00 Web Content
4331 pts/0    00:00:00 ps
```

---

5. Visualización del PID y nombre del proceso con `comm`

Se utilizó el siguiente comando:

```
ps -eo pid,comm
```

La opción `-e` permite seleccionar todos los procesos, mientras que `-o` permite definir las columnas que se desean visualizar.

En este caso se solicitaron:

- **PID:** identificador del proceso.
- **COMM:** nombre del comando o proceso.

La captura muestra una lista extensa de procesos del sistema, incluyendo procesos del kernel como `kworker`, servicios como `systemd`, aplicaciones como `firefox-esr` y procesos relacionados con GNOME.

Esta forma de mostrar los procesos facilita su identificación cuando únicamente se necesita conocer el PID y el nombre del proceso.

Capturas 9-12. Visualización de PID y COMM

```
user@debian:~$ ps -eo pid,comm
    PID COMMAND
      1 systemd
      2 kthreadd
      3 pool_workqueue_release
      4 kworker/R-kvfree_rcu_reclaim
      5 kworker/R-rcu_gp
      6 kworker/R-sync_wq
      7 kworker/R-slub_flushwq
      8 kworker/R-netns
      9 kworker/0:0-events
     10 kworker/0:0H-events_highpri
     13 kworker/R-mm_percpu_wq
     14 rcu_tasks_kthread
     15 rcu_tasks_rude_kthread
     16 rcu_tasks_trace_kthread
     17 ksoftirqd/0
     18 rcu_preempt
     19 rcu_exp_par_gp_kthread_worker/0
     20 rcu_exp_gp_kthread_worker
     21 migration/0
     22 idle_inject/0
     23 cpuhp/0
     24 cpuhp/1
     25 idle_inject/1
     26 migration/1
     27 ksoftirqd/1
     29 kworker/1:0H-events_highpri
     30 cpuhp/2
     31 idle_inject/2
     32 migration/2
     33 ksoftirqd/2
     35 kworker/2:0H-events_highpri
     36 cpuhp/3
     37 idle_inject/3
     38 migration/3
     39 ksoftirqd/3
     41 kworker/3:0H-events_highpri
     42 cpuhp/4
     43 idle_inject/4
     44 migration/4
     45 ksoftirqd/4
     47 kworker/4:0H-events_highpri
     48 cpuhp/5
     49 idle_inject/5
     50 migration/5
     51 ksoftirqd/5
     53 kworker/5:0H-events_highpri
     54 cpuhp/6
     55 idle_inject/6
     56 migration/6
     57 ksoftirqd/6
     58 kworker/6:0-mm_percpu_wq
     59 kworker/6:0H-events_highpri
```

```
 60 cpuhp/7
 61 idle_inject/7
 62 migration/7
 63 ksoftirqd/7
 65 kworker/7:0H-events_highpri
 69 kworker/u32:3-events_unbound
 73 kworker/u32:7-events_unbound
 74 kdevtmpfs
 75 kworker/R-inet_frag_wq
 76 kauditd
 77 khungtaskd
 78 oom_reaper
 79 kworker/R-writeback
 80 kcompactd0
 81 ksmd
 82 khugepaged
 83 kworker/R-kintegrityd
 84 kworker/R-kblockd
 85 kworker/R-blkcg_punt_bio
 86 irq/9-acpi
 87 kworker/2:1-events
 88 kworker/6:1-i915-unordered
 89 kworker/4:1-events
 90 kworker/1:1-events
 91 kworker/5:1-events
 92 kworker/R-tpm_dev_wq
 93 kworker/R-edac-poller
 94 kworker/R-devfreq_wq
 95 kworker/R-quota_events_unbound
 96 kworker/0:1H-events_highpri
 97 kswapd0
106 kworker/R-kthrotld
110 kworker/R-acpi_thermal_pm
111 hwrng
112 kworker/R-mld
113 kworker/R-ipv6_addrconf
118 kworker/R-kstrp
120 kworker/u33:0-rtw_tx_wq
175 kworker/5:1H-events_highpri
208 kworker/7:1H-events_highpri
209 kworker/7:2-events
218 kworker/2:1H-kblockd
227 kworker/6:1H-kblockd
228 kworker/1:1H-events_highpri
237 kworker/4:1H-events_highpri
256 kworker/3:1H-kblockd
259 kworker/1:2-events
262 watchdogd
263 irq/144-MSFT0001:01
264 kworker/R-nvme-wq
265 kworker/R-nvme-reset-wq
266 kworker/R-nvme-delete-wq
267 kworker/R-nvme-auth-wq
268 kworker/R-ata_sff
```

```
  285 card0-crtc1
  286 card0-crtc2
  287 card0-crtc3
  288 kworker/R-uas
  300 kworker/R-md
  301 kworker/R-md_bitmap
  307 kworker/R-raid5wq
  485 psimon
  499 systemd-journal
  547 kworker/2:2-events
  551 systemd-udevd
  556 psimon
  621 irq/172-mei_me
  637 kworker/R-cryptd
  638 kworker/R-cfg80211
  677 kworker/u32:10-events_unbound
  835 irq/173-rtw88_pci
  915 irq/174-AudioDSP
 1452 accounts-daemon
 1455 avahi-daemon
 1456 bluetoothd
 1457 cron
 1458 dbus-daemon
 1464 low-memory-moni
 1465 polkitd
 1466 smartd
 1467 switcheroo-cont
 1469 systemd-logind
 1470 udisksd
 1471 avahi-daemon
 1511 NetworkManager
 1512 wpa_supplicant
 1522 psimon
 1546 ModemManager
 1560 krfcommd
 1575 cupsd
 1593 gdm3
 1599 gdm-session-wor
 1607 systemd
 1609 (sd-pam)
 1629 dbus-daemon
 1631 pipewire
 1633 pipewire
 1634 wireplumber
 1635 pipewire-pulse
 1637 gnome-keyring-d
 1638 mpris-proxy
 1647 rtkit-daemon
 1650 gdm-wayland-ses
 1659 gnome-session-b
 1709 gcr-ssh-agent
 1710 gnome-session-c
 1711 ssh-agent
 1721 gvfsd
```

```
1729 gvfsd-fuse
1731 gnome-session-b
1765 at-spi-bus-laun
1783 dbus-daemon
1788 gnome-shell
1809 xdg-permission-
1817 cups-browsed
1840 at-spi2-registr
1841 colord
1877 gnome-shell-cal
1885 dconf-service
1888 evolution-sourc
1890 gjs
1910 upowerd
1911 ibus-daemon
1916 gsd-a11y-settin
1917 gsd-color
1918 gsd-datetime
1921 gsd-housekeepin
1924 gsd-keyboard
1926 gsd-media-keys
1928 gsd-power
1929 gsd-print-notif
1930 gsd-rfkill
1931 gsd-screensaver
1934 gsd-sharing
1936 gsd-smartcard
1937 gsd-sound
1939 gsd-usb-protect
1940 gsd-wacom
1957 gvfs-udisks2-vo
2014 gjs
2018 gvfs-mtp-volume
2043 gsd-printer
2048 gvfs-gphoto2-vo
2070 gsd-disk-utilit
2078 gnome-software
2081 evolution-alarm
2150 gvfs-goa-volume
2159 goa-daemon
2171 goa-identity-se
2174 gvfs-afc-volume
2185 Xwayland
2249 xdg-desktop-por
2254 xdg-document-po
2257 evolution-calen
2263 fusermount3
2285 gvfsd-metadata
2292 gsd-xsettings
2297 evolution-addre
2321 mutter-x11-fram
2392 ibus-dconf
2393 ibus-extension-
2399 ibus-portal
```

```
2403 xdg-desktop-por
2422 ibus-engine-moz
2430 mozc_server
2465 xdg-desktop-por
2506 fwupd
2592 ibus-engine-sim
2599 power-profiles-
2707 firefox-esr
2719 crashhelper
2795 Socket Process
2856 WebExtensions
2872 RDD Process
2895 Privileged Cont
2950 Isolated Web Co
3027 Utility Process
3041 Isolated Web Co
3112 kworker/u33:3-i915_flip
3186 dbus
3187 dbus
3582 Web Content
3630 Web Content
3839 kworker/1:0-events
3849 kworker/0:1-mm_percpu_wq
3869 kworker/4:0
3919 gnome-terminal-
3927 bash
3958 kworker/5:0-mm_percpu_wq
4022 kworker/7:0-i915-unordered
4023 Web Content
4056 kworker/3:1-events
4182 kworker/2:0-events
4342 kworker/5:2
4352 ps
```

---

6. Uso de CPU

Finalmente, se utilizó el comando:

```
ps -eo pid,comm,%cpu --sort=-%cpu
```

Este comando permite mostrar el **PID**, el nombre del proceso (**COMM**) y el porcentaje de **CPU** utilizado.

La opción:

```
--sort=-%cpu
```

ordena los resultados de mayor a menor porcentaje de utilización de CPU.

En la captura se observa que el proceso con mayor consumo en ese momento era:

- **PID 3041 – Isolated Web Content:** aproximadamente **30.3 % de CPU**.

- **PID 2707 – firefox-esr:** aproximadamente **12.0 % de CPU**.
- **PID 1788 – gnome-shell:** aproximadamente **3.0 % de CPU**.
- **PID 2078 – gnome-software:** aproximadamente **1.4 % de CPU**.

Esto permite identificar rápidamente qué procesos están utilizando una mayor cantidad del procesador. En este caso, los procesos relacionados con Firefox presentaron un consumo considerable debido a la actividad del navegador.

Es importante mencionar que el porcentaje de CPU puede cambiar constantemente dependiendo de las aplicaciones abiertas y de las tareas que esté realizando el sistema.

Captura 13. Procesos ordenados por consumo de CPU

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```
user@debian:~$ ps -eo pid,comm,%cpu --sort=-%cpu
    PID COMMAND         %CPU
   3041 Isolated Web Co 30.3
   2707 firefox-esr     12.0
   1788 gnome-shell      3.0
   2078 gnome-software   1.4
    835 irq/173-rtw88_p  0.9
   3919 gnome-terminal-  0.9
   2506 fwupd            0.3
   2393 ibus-extension-  0.2
   2895 Privileged Cont  0.2
     69 kworker/u32:3-e  0.1
      1 systemd          0.1
   1911 ibus-daemon      0.1
   1511 NetworkManager   0.1
   2950 Isolated Web Co  0.1
   1458 dbus-daemon      0.0
   2081 evolution-alarm  0.0
   4023 Web Content      0.0
    677 kworker/u32:10-  0.0
   1888 evolution-sourc  0.0
    282 usb-storage      0.0
   3630 Web Content      0.0
   3582 Web Content      0.0
    120 kworker/u33:0-r  0.0
   2185 Xwayland         0.0
   2856 WebExtensions    0.0
   1634 wireplumber      0.0
     18 rcu_preempt      0.0
     73 kworker/u32:7-a  0.0
   4438 kworker/u32:0-e  0.0
   2402 localsearch-3    0.0
   2430 mozc_server      0.0
   1512 wpa_supplicant   0.0
   3112 kworker/u33:3-i  0.0
    209 kworker/7:2-eve  0.0
   4022 kworker/7:0-i91  0.0
   2422 ibus-engine-moz  0.0
   1629 dbus-daemon      0.0
   1631 pipewire         0.0
   1607 systemd          0.0
   1465 polkitd          0.0
   2872 RDD Process      0.0
    499 systemd-journal  0.0
   2297 evolution-addre  0.0
   1926 gsd-media-keys   0.0
   1928 gsd-power        0.0
     88 kworker/6:1-eve  0.0
     82 khugepaged       0.0
   2257 evolution-calen  0.0
   1940 gsd-wacom        0.0
   1841 colord           0.0
   1635 pipewire-pulse   0.0
   1924 gsd-keyboard     0.0
```

```
1455 avahi-daemon     0.0
3027 Utility Process  0.0
  57 ksoftirqd/6      0.0
2321 mutter-x11-fram  0.0
2249 xdg-desktop-por  0.0
1910 upowerd          0.0
1470 udisksd          0.0
1546 ModemManager     0.0
1469 systemd-logind   0.0
2292 gsd-xsettings    0.0
1917 gsd-color        0.0
  91 kworker/5:1-eve  0.0
 274 kworker/3:2-mm_  0.0
  89 kworker/4:1-mm_  0.0
1877 gnome-shell-cal  0.0
1731 gnome-session-b  0.0
1659 gnome-session-b  0.0
1452 accounts-daemon  0.0
  63 ksoftirqd/7      0.0
3927 bash             0.0
1934 gsd-sharing      0.0
3849 kworker/0:1-eve  0.0
3839 kworker/1:0-eve  0.0
4182 kworker/2:0-eve  0.0
2465 xdg-desktop-por  0.0
2014 gjs              0.0
1637 gnome-keyring-d  0.0
1456 bluetoothd       0.0
 915 irq/174-AudioDS  0.0
2795 Socket Process   0.0
 278 kworker/0:2-rcu  0.0
 208 kworker/7:1H-ev  0.0
  17 ksoftirqd/0      0.0
  26 migration/1      0.0
  27 ksoftirqd/1      0.0
  33 ksoftirqd/2      0.0
2599 power-profiles-  0.0
2174 gvfs-afc-volume  0.0
2159 goa-daemon       0.0
1921 gsd-housekeepin  0.0
1817 cups-browsed     0.0
1593 gdm3             0.0
1575 cupsd            0.0
1466 smartd           0.0
  90 kworker/1:1-eve  0.0
  32 migration/2      0.0
  38 migration/3      0.0
  39 ksoftirqd/3      0.0
  44 migration/4      0.0
  45 ksoftirqd/4      0.0
  50 migration/5      0.0
  51 ksoftirqd/5      0.0
  56 migration/6      0.0
  62 migration/7      0.0
```

```
1957 gvfs-udisks2-vo  0.0
1937 gsd-sound        0.0
1890 gjs              0.0
2254 xdg-document-po  0.0
1936 gsd-smartcard    0.0
1721 gvfsd            0.0
1633 pipewire         0.0
 237 kworker/4:1H-kb  0.0
 227 kworker/6:1H-ev  0.0
 218 kworker/2:1H-ev  0.0
  96 kworker/0:1H-ev  0.0
  80 kcompactd0       0.0
2592 ibus-engine-sim  0.0
2171 goa-identity-se  0.0
1929 gsd-print-notif  0.0
1647 rtkit-daemon     0.0
1599 gdm-session-wor  0.0
1464 low-memory-moni  0.0
 259 kworker/1:2-eve  0.0
 256 kworker/3:1H-ev  0.0
 228 kworker/1:1H-ev  0.0
 175 kworker/5:1H-ev  0.0
2285 gvfsd-metadata   0.0
2150 gvfs-goa-volume  0.0
2070 gsd-disk-utilit  0.0
2048 gvfs-gphoto2-vo  0.0
2043 gsd-printer      0.0
2018 gvfs-mtp-volume  0.0
1939 gsd-usb-protect  0.0
1930 gsd-rfkill       0.0
1931 gsd-screensaver  0.0
1916 gsd-a11y-settin  0.0
1918 gsd-datetime     0.0
1885 dconf-service    0.0
1840 at-spi2-registr  0.0
1711 ssh-agent        0.0
1638 mpris-proxy      0.0
1467 switcheroo-cont  0.0
1457 cron             0.0
 111 hwrng            0.0
   9 kworker/0:0-eve  0.0
  21 migration/0      0.0
  58 kworker/6:0-mm_  0.0
   2 kthreadd         0.0
   3 pool_workqueue_  0.0
   4 kworker/R-kvfre  0.0
   5 kworker/R-rcu_g  0.0
   6 kworker/R-sync_  0.0
   7 kworker/R-slub_  0.0
   8 kworker/R-netns  0.0
  10 kworker/0:0H-ev  0.0
  13 kworker/R-mm_pe  0.0
  14 rcu_tasks_kthre  0.0
  15 rcu_tasks_rude_  0.0
```

```
    20 rcu_exp_gp_kthr  0.0
    22 idle_inject/0    0.0
    23 cpuhp/0          0.0
    24 cpuhp/1          0.0
    25 idle_inject/1    0.0
    29 kworker/1:0H-ev  0.0
    30 cpuhp/2          0.0
    31 idle_inject/2    0.0
    35 kworker/2:0H-ev  0.0
    36 cpuhp/3          0.0
    37 idle_inject/3    0.0
    41 kworker/3:0H-ev  0.0
    42 cpuhp/4          0.0
    43 idle_inject/4    0.0
    47 kworker/4:0H-ev  0.0
    48 cpuhp/5          0.0
    49 idle_inject/5    0.0
    53 kworker/5:0H-ev  0.0
    54 cpuhp/6          0.0
    55 idle_inject/6    0.0
    59 kworker/6:0H-ev  0.0
    60 cpuhp/7          0.0
    61 idle_inject/7    0.0
    65 kworker/7:0H-ev  0.0
    74 kdevtmpfs        0.0
    75 kworker/R-inet_  0.0
    76 kauditd          0.0
    77 khungtaskd       0.0
    78 oom_reaper       0.0
    79 kworker/R-write  0.0
    81 ksmd             0.0
    83 kworker/R-kinte  0.0
    84 kworker/R-kbloc  0.0
    85 kworker/R-blkcg  0.0
    86 irq/9-acpi       0.0
    87 kworker/2:1-eve  0.0
    92 kworker/R-tpm_d  0.0
    93 kworker/R-edac-  0.0
    94 kworker/R-devfr  0.0
    95 kworker/R-quota  0.0
    97 kswapd0          0.0
   106 kworker/R-kthro  0.0
   110 kworker/R-acpi_  0.0
   112 kworker/R-mld    0.0
   113 kworker/R-ipv6_  0.0
   118 kworker/R-kstrp  0.0
   262 watchdogd        0.0
   263 irq/144-MSFT000  0.0
   264 kworker/R-nvme-  0.0
   265 kworker/R-nvme-  0.0
   266 kworker/R-nvme-  0.0
   267 kworker/R-nvme-  0.0
   268 kworker/R-ata_s  0.0
   269 irq/143-GTCH750  0.0
```

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```
    270 scsi_eh_0        0.0
    271 kworker/R-scsi_  0.0
    272 scsi_eh_1        0.0
    273 kworker/R-scsi_  0.0
    277 spi0             0.0
    280 scsi_eh_2        0.0
    281 kworker/R-scsi_  0.0
    283 kworker/R-ttm    0.0
    284 card0-crtc0      0.0
    285 card0-crtc1      0.0
    286 card0-crtc2      0.0
    287 card0-crtc3      0.0
    288 kworker/R-uas    0.0
    300 kworker/R-md     0.0
    301 kworker/R-md_bi  0.0
    307 kworker/R-raid5  0.0
    485 psimon           0.0
    556 psimon           0.0
    621 irq/172-mei_me   0.0
    637 kworker/R-crypt  0.0
    638 kworker/R-cfg80  0.0
   1471 avahi-daemon     0.0
   1522 psimon           0.0
   1560 krfcommd         0.0
   1609 (sd-pam)         0.0
   1650 gdm-wayland-ses  0.0
   1709 gcr-ssh-agent    0.0
   1710 gnome-session-c  0.0
   1729 gvfsd-fuse       0.0
   1765 at-spi-bus-laun  0.0
   1783 dbus-daemon      0.0
   1809 xdg-permission-  0.0
   2263 fusermount3      0.0
   2392 ibus-dconf       0.0
   2399 ibus-portal      0.0
   2719 crashhelper      0.0
   3186 dbus             0.0
   3187 dbus             0.0
   3869 kworker/4:0-eve  0.0
   3958 kworker/5:0-mm_  0.0
   4056 kworker/3:1-eve  0.0
   4342 kworker/5:2      0.0
   4355 kworker/3:0      0.0
   4357 kworker/4:2-eve  0.0
   4433 kworker/7:1      0.0
   4437 kworker/u33:1    0.0
   4462 ps               0.0
user@debian:~$
```

Resultados obtenidos

A partir de los comandos utilizados fue posible identificar diferentes características de los procesos de Linux:

Comando	Función principal
`ps`	Muestra los procesos asociados a la terminal actual.
`ps -ef`	Muestra información detallada de todos los procesos.
`ps -ef | grep firefox`	Filtra los procesos relacionados con Firefox.
`ps -u user`	Muestra los procesos pertenecientes al usuario indicado.
`ps -eo pid,comm`	Muestra el PID y nombre de cada proceso.
`ps -eo pid,comm,%cpu --sort=-%cpu`	Muestra los procesos junto con su consumo de CPU, ordenados de mayor a menor.

Durante la práctica se comprobó que un sistema Linux mantiene una gran cantidad de procesos activos, muchos de ellos correspondientes a servicios internos, procesos del kernel y componentes del entorno gráfico.

También se comprobó que una aplicación como Firefox puede utilizar varios procesos simultáneamente. Esto permite distribuir diferentes tareas y mejorar la estabilidad y seguridad de la aplicación.

Conclusión

La realización de esta práctica permitió comprender de manera básica cómo Linux administra y muestra los procesos que se encuentran activos en el sistema. Mediante el comando `ps` fue posible obtener información general, mientras que `ps -ef` permitió observar información más detallada como el usuario, PID, proceso padre y comando.

El uso de `grep` facilitó la búsqueda de procesos específicos, en este caso los relacionados con Firefox. Por otra parte, `ps -u user` permitió identificar los procesos pertenecientes a un usuario determinado, mientras que las opciones `-eo` permitieron seleccionar específicamente la información que se deseaba visualizar.

Finalmente, mediante el análisis del porcentaje de CPU fue posible identificar cuáles procesos estaban utilizando más recursos del procesador. Con esto se comprendió que los comandos de Linux para la administración de procesos son herramientas importantes para **monitorear el rendimiento, identificar aplicaciones activas y conocer el uso de los recursos del sistema**.

En conclusión, la práctica permitió adquirir una base para el manejo y análisis de procesos en Linux mediante la terminal, fortaleciendo el conocimiento sobre la estructura y funcionamiento del sistema operativo.